EE1100: Basic Electrical Engineering

Tutorial 4: Resonance, Filter and Three-Phase Circuits

Q1. A series RLC circuit has the following parameter values: R= 10 Ω, L= 0.01 H, C= 100 μF.

(a) Compute the resonant frequency in radians per second.
(b) Calculate the quality factor of the circuit.
(c) What is the value of the bandwidth?
(d) Compute the lower and upper frequency points of the bandwidth.
(e) If a signal of e(t) = 1 sin 1000t is applied to this series RLC circuit, calculate the maximum value of voltage appearing across the capacitor terminals.

[Ans.: (a) 1000 rad/s (b) 1 (c) 1000 rad/s (d) 618.03 rad/s, 1618.03 rad/s (e) 1 V]

Q2. A current source is applied to the parallel arrangement of R, L and C where R=10 Ω, L=1 H and C=1 μF.

(a) Compute the resonant frequency.
(b) Find the quality factor.
(c) Calculate the value of the bandwidth.
(d) Compute the lower and upper frequency points of the bandwidth.
(e) If a signal of i(t) = 1 sin 1000t is applied to this parallel RLC circuit, calculate the maximum value of voltage appearing across the capacitor terminals?
(f) What is the capacitor current in part (e)?

[Ans.: (a) 1000 rad/s (b) 0.01 (c) 100000 rad/s (d) 10 rad/s, 100010 rad/s (e) 10 V (f) $0.00707\angle 90^0$ A]

Q3. A voltage of e(t) = 10 sin ωt is applied to a series RLC circuit. At the resonant frequency of the circuit the maximum voltage across the capacitor is found to be 500 V. Moreover, the bandwidth is known to be 400 rad/sec and the impedance at resonant is 100 Ω.

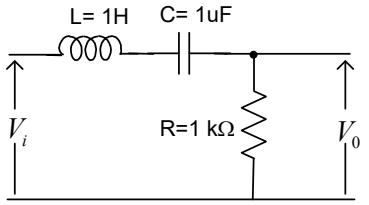
(a) Find the resonant frequency?
(b) Compute the upper and lower limits of the bandwidth?
(c) Determine the value of L and C for this circuit?

[Ans.: (a) 20000 rad/s (b) 19800 rad/s, 20200 rad/s (c) 0.25 H, 0.01 μF]

Q4. A source generates a low frequency signal that contains noise. It appears that the noise could be removed with a simple low pass filter consisting of resistance and capacitor elements, with a half power frequency of 1 kHz. If 1.6 μF capacitor is available, select the resistor to eliminate the noise. [Ans.: 100 Ω]

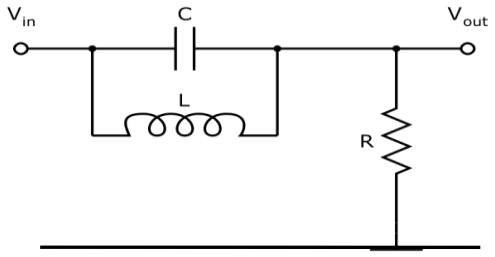
Q5. Design a high pass filter consisting of resistance (1 kΩ) and capacitor elements, with break frequency or lower cut-off frequency as 20 kHz. [Ans.: 7.95 nF].

Q6. (a) Consider the following circuit. Find out its transfer function and the nature of filter characteristics. Determine the center frequency, half power frequencies, and bandwidth.



[Ans.: 1000 rad/s, 618 rad/s, 1818 rad/s, 1000 rad/s]

Q7. The internal communication is taking place in small plane. But there is an interference at 400 Hz, which is aircraft power system frequency. If the speaker is modelled as 4 kΩ resistor, design a band rejection filter to be placed at the input of the speaker system to eliminate 400 Hz frequency component as shown in the fig. Consider capacitance value of filter as 200 μF.
[Ans.: L= 791.6 μH]



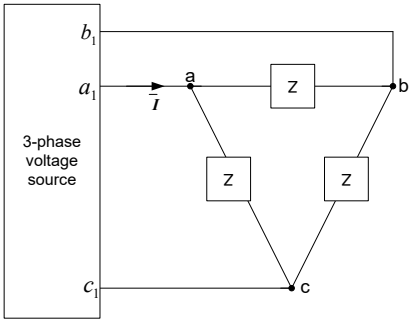
Q8. A university student is designing a simple FM receiver to tune to the campus radio station at 100 MHz. Two nearby stations operate at 98 MHz and 102 MHz. The receiver front-end uses a series RLC band-pass filter with the output taken across a 100 Ω resistor. The filter should provide maximum response at 100 MHz and attenuate each neighbouring station by at least 20 dB relative to the desired station.

Using 100 Ω resistor, a variable inductor, and a variable capacitor, design the filter and determine:
(a) the minimum required quality factor Q,
(b) the corresponding values of L and C, and
(c) comment briefly on whether the obtained component values are practically realizable for an FM receiver.

[Ans.: (a) Qmin = 251.2 (b) L = 39.98 μH, C = 0.0634 pF (c) Theoretically realizable, but difficult to implement as a simple lumped RLC filter at 100 MHz because of the high Q and parasitic effects.]

Q9. A 220-V (L-L), three phase balanced voltage is applied to a balanced delta-connected three phase load in the manner illustrated in the figure below. The phase current measured between points *a* and *b* is $\bar{I}_{ab} = 10\angle -30^0$ A.

(a) Find the phase and line currents**.** Draw the phasor diagram showing clearly the line voltages, phase currents and line currents.
(b) Compute the total power received by the three-phase load.
(c) Find the value of the resistive portion of the phase impedance.



[Ans.: (a) $10\angle -30^0$ A, $10\angle -150^0$ A, $10\angle 90^0$ A, $17.32\angle -60°$ A, $17.32\angle -180°$ A, $17.32\angle -300°$ A (b) 5715.6 W, 2857.8 Var (lag) (c) 19.05 Ω

Q10. The delta-connected load of Fig. 1 consists of phase impedance each equal to $15 + j20\ \Omega$.

(a) Find the phasor current in each line.

(b) What is the power consumed per phase?

(c) What is the phasor sum of the three phase line currents? Why does it have this value?

[Ans.: (a) $8.8\angle -53.13^0$, $8.8\angle -173.13^0$, $8.8\angle 66.86^0$ A (b) 1161.6 W, 1548.8 Var (lag) (c) 0 A

Q11. A three-phase, 208 V generator supplies a total 1800 W at a line current of 10 A. When three identical impedances are arranged in a star connection across the line terminals of the generator. Compute the resistive and reactive components of each phase impedance.

[Ans.: 6 Ω, ±10.39 Ω]

Q12. A balanced three-phase star-connected load has an impedance of $4\angle 60^0\ \Omega$ from line to neutral. Moreover, from line *a* to neutral *n* the voltage in $\overline{V}_{an} = 20\angle 30^0$ V.

(a) What is the current in phases a, *b* and *c*?

(b) What is the voltage from line *b* to neutral *n*?

(c) What is the phasor expression for the voltage from line *a* to line *c* , i.e., $\overline{V}_{ac}$?

[Ans.: (a) $5\angle -30^0$, $5\angle -150^0$, $5\angle 90^0$ A (b) $20\angle -90^0$ V (c) $34.64\ \angle 0^0$ V]